

# Lab: Explore Prompt Writing



Estimated Time: **15** minutes

## Objectives

- Compare and contrast prompt writing approaches
- Choose a prompt approach based on the goal of your interaction with the AI
- Write a prompt using key elements for writing effective prompts and the prompt writing process

## Prerequisites

You should be familiar with:

- Elements of good prompts
- Prompt writing process
- Approaches to writing prompts

## Introduction

In this lab, you will explore writing different prompts based on a scenario we provide. We will provide you with prompt choice elements. You will mix and match these prompt elements to form similar but varying prompts so you can compare and contrast outputs based on the elements you chose.

# Scenario

A director of a small non-profit organization dedicated to working with its community to improving the longevity and quality of life of its members. They want to launch a new program educating seniors on how to sensibly use AI. She has heard the term "AI hallucinations" and wants to understand the term better to see if it's the program needs to address the topic.

# Review

You can choose to skip this **Review** section and continue to the **Practice** section if you feel comfortable with the prerequisites.

## Elements of good prompts

- Define a clear goal for the interaction
- Avoid ambiguity, write clearly and concisely
- Provide context
  - Specify output expectations:
  - Specify desired output structure
  - Identify constraints for the output
  - Provide a point of view / persona
  - Ask the AI to state its assumptions

# Prompt writing approaches

## Interaction

| Purpose                   | Description                                        | Approaches                                          |
|---------------------------|----------------------------------------------------|-----------------------------------------------------|
| Information retrieval     | Factual knowledge                                  | Zero-shot, One-shot, Few-shot, Interview            |
| Analysis and reflection   | Insight generation, exploration, logical analysis  | Chain-of-thought, Self-reflection, Tree-of-thought  |
| Creative idea generation  | Brainstorming                                      | Role-play, Scenario simulation                      |
| Instructional guidance    | Sequential, structured output                      | Process decomposition, Templating, Guided narrative |
| Collaborative exploration | Open-ended, unpacking unfamiliar or complex topics | Conversational prompting, Iterative refinement      |

## Prompt writing process

1. Define your goal, including the purpose of your interaction. Your purpose will help you decide on which approach(es) will help you meet your goal.
2. Write and test the prompt
3. Analyze the response
4. Iterate

# Practice

In this guided practice exercise, you will explore writing different prompts to learn more about the term "AI hallucination" from the perspective of the scenario provided at the beginning of the lab.

## Recall the scenario

A director of a small non-profit organization dedicated to working with its community to improving the longevity and quality of life of its members. They want to launch a new program educating seniors on how to sensibly use AI. She has heard the term "AI hallucinations" and wants to understand the term better to see if the program needs to address the topic.

## Instructions

*In the generative AI tool, select the GPT-5 Mini model at the top of the window. You do not need to enter any prompt instructions.*

You will construct prompts from prompt elements we provide you. You will mix and match to compare outputs from similar, but varying, prompts.

### Step 1. Goal choice

Topic: AI hallucinations Goal: Determine the depth to which the topic of AI hallucinations needs addressed in the provided scenario.

### Step 2. Write and test the prompt

In this step, you will see:

- Sample prompt questions

- Output expectations
1. Choose one element from each section to construct your prompt, then enter it into the *AI Classroom* tool.
  2. Copy and paste the prompts and the responses into a text document so you can compare them.

## Sample prompts (Choose one)

- Define the term "AI hallucination." (Information retrieval)
- Why do AI models hallucinate? (Analysis)
- Provide four examples of widely reported hallucinations. Describe the factors that made these hallucinations particularly noteworthy. (Analysis, information retrieval)
- What contexts would motivate seniors to learn the term "AI hallucinations"? (Idea generation)
- What causes hallucinations? Explain the steps you used to reach this conclusion. (Instructional guidance)
- Why would a senior community member need to understand the term "AI hallucination"? (Collaborative exploration)

## Possible contexts (Choose one)

- I'm developing a curriculum to educate seniors in my community on how to use AI so they can use AI wisely.
- I'm working with my community to improve the longevity and quality of life of its members.
- As a senior consumer wanting to use AI, I want to better understand the term "AI hallucinations".

## Output expectations (Choose at least one)

- Explain why you chose this approach
- Identify potential assumptions
- Provide your references

## Example prompt

- Sample prompt 3
- Context 1
- Output expectation 4

*Provide four examples of widely reported hallucinations. Describe the factors that made these hallucinations particularly noteworthy. I'm developing a curriculum to educate seniors in my community on how to use AI so they can use AI wisely so I would like to use these factors to help them understand why this is an important concept when using AI. Provide your references.*

**Note:** If this prompt gives error, you can split the original prompt into two sequential prompts as given below:

Prompt 1: Provide four examples of widely reported AI hallucinations and briefly describe what happened in each case. I'm developing a curriculum to educate seniors in my community on how to use AI so they can use AI wisely.

Prompt 2: Describe the factors that made these hallucinations particularly noteworthy and explain why these factors are important for seniors to understand when learning to use AI wisely. Provide your references.

Save the responses generated from both prompts. Combine the outputs from Prompt 1 and Prompt 2 to review the complete response for the activity.

## Step 3. Analyze

Evaluate how well the output aligns with your goal.

Consider:

- Is the response free from bias?
- Do you need a more conceptual response or a specific response?
- Do you want a different point of view?

## Step 4. Iterate

After considering the output, either revise and refine the entire prompt and try again or further investigate an aspect of the output.

## For further exploration

### Choose your own topic to investigate with AI

- Decide on your goal
- Write and test the prompt
- Word your inquiry using concise phrasing while also providing sufficient information to achieve your objective
- Provide appropriate context
- Specify your output expectations, including structure, constraints, persona, and bias considerations for the output

## Evaluate

Consider how well the output aligns with your goal:

Is there any information missing

Is the response free from bias?

Do you need a more conceptual response or a specific response?

Do you want a different point of view?

## Iterate

After considering the output, either revise and refine the entire prompt and try again or



# Congratulations!

You have completed this lab. You now have experience with:

- Comparing and contrasting prompt writing approaches
- Choosing a prompt approach based on the goal of your interaction with the AI
- Writing a prompt using key elements for writing effective prompts and the prompt writing process

## Next Steps

You can apply the prompt techniques from this lab to real world needs. Return to this lab if needed to revisit these techniques.

## Author

Bethany Hudnutt

© IBM Corporation. All rights reserved.

# Lab: Explore Prompt Writing



Estimated Time: **15** minutes

# Objectives

- Compare and contrast prompt writing approaches
- Choose a prompt approach based on the goal of your interaction with the AI
- Write a prompt using key elements for writing effective prompts and the prompt writing process

## Prerequisites

You should be familiar with:

- Elements of good prompts
- Prompt writing process
- Approaches to writing prompts

## Introduction

In this lab, you will explore writing different prompts based on a scenario we provide. We will provide you with prompt choice elements. You will mix and match these prompt elements to form similar but varying prompts so you can compare and contrast outputs based on the elements you chose.

## Scenario

A director of a small non-profit organization dedicated to working with its community to improving the longevity and quality of life of its members. They want to launch a new program educating seniors on how to sensibly use AI. She has heard the term "AI hallucinations" and wants to understand the term better to see if it's the program needs to address the topic.

Review →